IGCSE 0607 Practice Questions — E4.2 — Combined Questions & Markscheme

Gradient of linear graphs

Total Questions: 18 — **Total Marks:** 107 Marks

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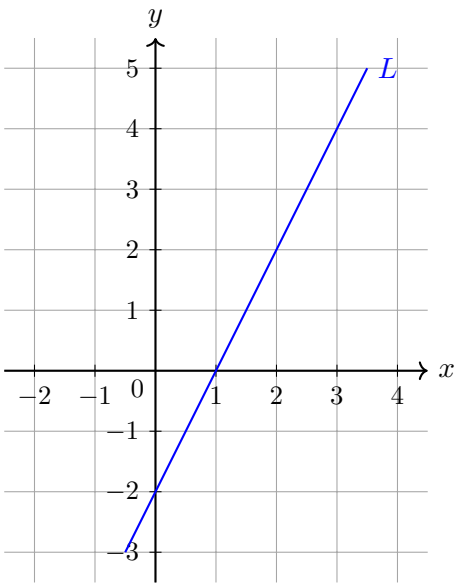
- **E4.2** Gradient of linear graphs
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 Points A and B have coordinates $(2, -3)$ and $(6, 5)$ respectively.

Work out the gradient of the line AB.

..... **[2]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the gradient of line L.

..... **[2]**

(b) Write down the coordinates of the y-intercept of line L.

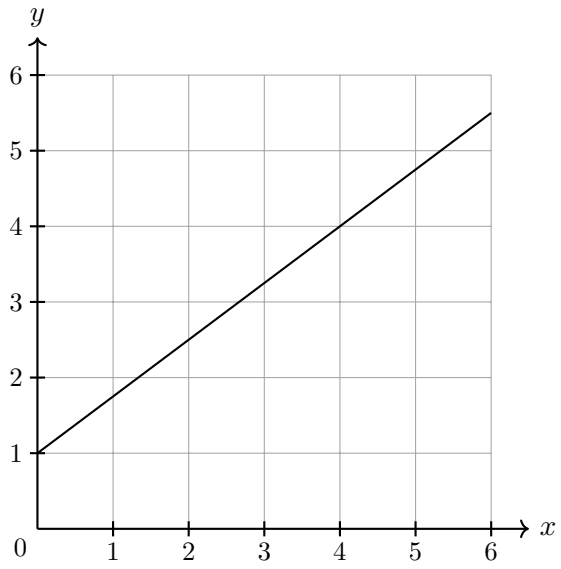
(..... , ...) **[1]**

3 The points A and B have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

(a) Work out the gradient of the line AB.

..... **[2]**

(b) A straight line L is drawn on the grid.
Find the gradient of line L.



..... **[2]**

4 The points $A(-4, 7)$ and $B(2, -5)$ lie on a straight line.

(a) Calculate the gradient of the line AB.

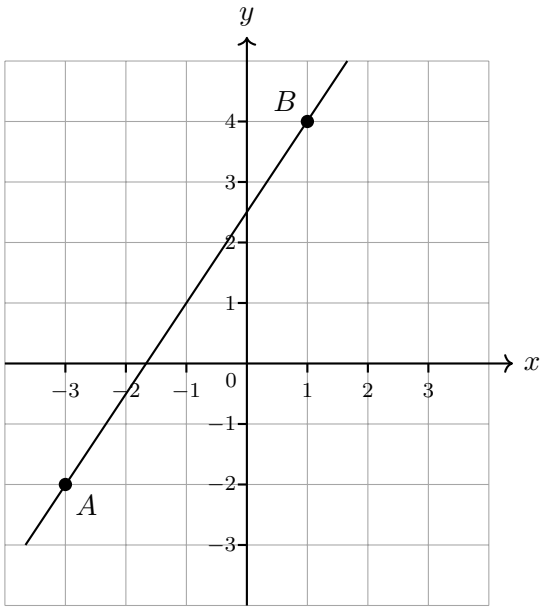
..... **[2]**

(b) Point C has coordinates $(0, c)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of c.

$c =$ ..... **[2]**

5 **(a)** The diagram shows a straight line passing through points A and B.

Work out the gradient of the line AB.



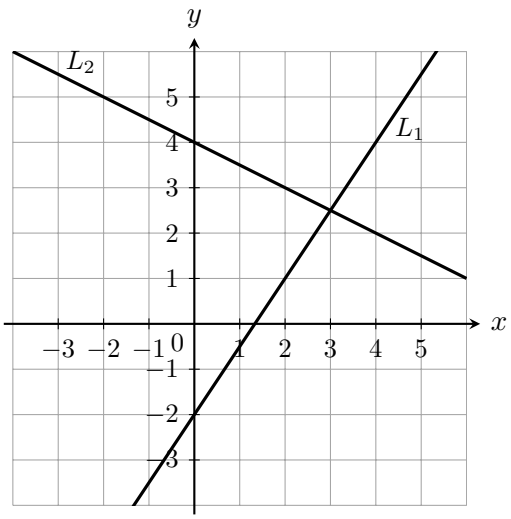
Gradient = ... **[2]**

(b) Point C has coordinates $(-2, 5)$ and point D has coordinates $(4, -7)$.

Calculate the gradient of the line CD.

Gradient = ... **[2]**

6 The diagram shows two straight lines, L_1 and L_2, drawn on a coordinate grid.



(a) Find the gradient of line L_1.

..... **[2]**

(b) Find the gradient of line L_2.

..... **[2]**

(c) A straight line L_3 passes through the points $P(-3, 7)$ and $Q(5, -9)$.
Calculate the gradient of line L_3.

..... **[2]**

7 **(a)** A straight line passes through the points $P(2, k)$ and $Q(5, 3k - 1)$.

(i) Find an expression for the gradient of the line PQ in terms of k.
Give your answer in its simplest form.

Gradient = ... **[2]**

(ii) The gradient of the line PQ is 4.
Find the value of k.

k = ... **[2]**

8 The points $A(-2, 5)$, $B(4, k)$, and $C(10, -1)$ lie in a coordinate plane.

(a) Given that the gradient of the line segment AB is $-\frac{1}{2}$, calculate the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), show that the points A, B, and C lie on the same straight line.

[2]

(c) Point D has coordinates $(p, 3p - 1)$.
Find the value of p for which the line AD has a gradient of 2.

$p =$ ..... **[3]**

9 **(a)** Point P has coordinates $(2a, 3a)$ and point Q has coordinates $(5a, a + 4)$.

Find an expression for the gradient of the line PQ in terms of a.
Give your answer in its simplest form.

..... **[2]**

(b) The line PQ is perpendicular to a line with gradient 3.
Work out the value of a.

$a =$ ..... **[3]**

(c) The points $A(-1, 2)$, $B(3, 8)$ and $C(c, 23)$ lie on a straight line.
Work out the value of c.

$c =$ ..... **[3]**

10 The line L_1 passes through the points $P(-2, 3a)$ and $Q(4, a+6)$.

(a) Find an expression for the gradient of L_1 in terms of a.
Give your answer in its simplest form.

..... **[3]**

(b) The line L_2 has equation $y = 3x - 5$.
Given that L_1 is parallel to L_2, find the value of a.

$a =$ ..... **[2]**

(c) Using your value of a, find the coordinates of the midpoint of PQ.

(..... , ...) **[3]**

11 The points $A(1, 2)$, $B(5, k)$ and $C(k, 2k)$ are three distinct collinear points, where k is a constant.

(a) Write down an expression, in terms of k, for the gradient of the line segment AB.

..... **[1]**

(b) By finding an expression for the gradient of BC, show that $k^2 - 11k + 10 = 0$.

[3]

(c) Solve the equation to find the two possible values of k.

$k =$ ..... , k =

..... **[2]**

(d) Calculate the exact gradient of the line passing through point B and the origin $(0, 0)$.

..... **[2]**

12 The points A, B and C have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

(a) The points A, B and C lie on the same straight line.
Work out the value of y.

$y =$ ..... **[2]**

(b) A different point D has coordinates $(p, p^2 - p - 3)$.
The gradient of the line AD is 2.
Work out the possible values of p.

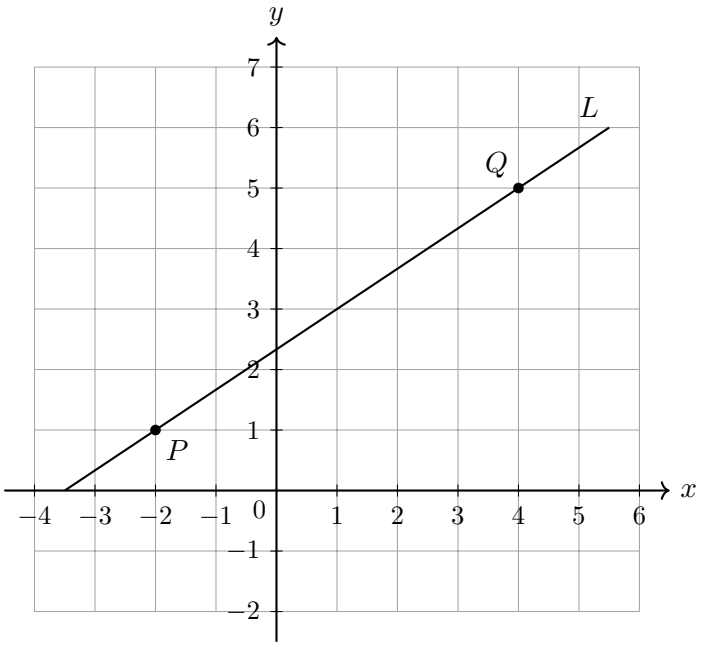
$p =$ ..... , $p =$ ..... **[4]**

(c) A line L has gradient m.
The line L passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

13 **(a)** The diagram shows a straight line L passing through the points $P(-2, 1)$ and $Q(4, 5)$.

Find the gradient of line L.



..... **[2]**

(b) A different line passes through the points $(3, k)$ and $(8, -2)$.

The gradient of this line is -1.4.

Calculate the value of k.

$k =$ ..... **[3]**

14 Point A has coordinates $(-3, 5)$ and point B has coordinates $(4, -9)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(k, 1)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of k.

$k =$ ..... **[3]**

15 Point P has coordinates $(k-3, 2k)$ and point Q has coordinates $(2, 5)$.

(a) The gradient of the line PQ is -3.
Work out the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), find the coordinates of the midpoint of PQ.

(..... , ...) **[2]**

16 A straight line passes through the points $P(3k, 4)$ and $Q(k-2, 8)$.
The gradient of this line is $-\frac{1}{2}$.

(a) Find the value of k.

$k =$ ..... **[3]**

(b) Write down the coordinates of P and Q.

$P =$ ..... , $Q =$ ..... **[1]**

(c) Another point R lies on the y-axis such that the gradient of the line passing through P and R is 1.
Find the coordinates of R.

(..... , ...) **[3]**

17 Points $A(-1, -4)$ and $B(3, 8)$ lie on a straight line.

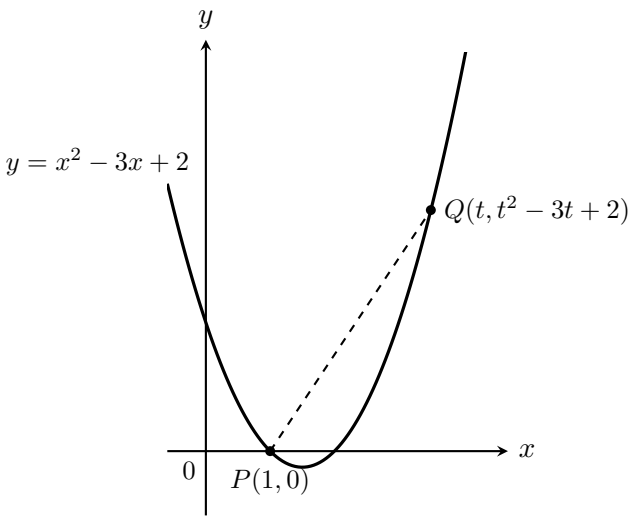
(a) Point $C(p, 23)$ also lies on the line passing through A and B.
Find the value of p.

$p =$ ..... **[3]**

(b) Point D has coordinates $(q, 2q - 1)$.
The gradient of the line AD is equal to $-\frac{1}{3}$.
Find the value of q.

$q =$ ..... **[3]**

18 The diagram shows a sketch of the curve $y = x^2 - 3x + 2$ and two points, $P(1, 0)$ and $Q(t, t^2 - 3t + 2)$, on the curve, where $t > 1$.



(a) Show that the gradient of the chord PQ simplifies to $t - 2$.

[3]

(b) Find the coordinates of point Q when the gradient of the chord PQ is 2.8.

$Q = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

(c) Point R has coordinates $(3.2, -1.5)$.
Find the value of t, where $t > 1$, for which the gradient of QR is equal to the gradient of PQ.
Give your answer correct to 3 significant figures.

$t =$ ..... **[4]**

1 $\frac{5-(-3)}{6-2}$ or equivalent **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** Evidence of using $\frac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\frac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** Uses gradient formula: $\frac{-4-8}{4-(-2)}$ or $\frac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\frac{3}{4}$ or 0.75 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Uses the gradient formula: $\frac{-5-7}{2-(-4)}$ **[M1]**

-2 **[A1]**

(b) Sets up the equation: $\frac{c-7}{0-(-4)} = -2$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Uses difference in y over difference in x, e.g., $\frac{4-(-2)}{1-(-3)}$ **[M1]**

$\frac{3}{2}$ or 1.5 **[A1]**

(b) $\frac{-7-5}{4-(-2)}$ or $\frac{5-(-7)}{-2-4}$ **[M1]**

-2 **[A1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** Correct vertical and horizontal change seen, e.g. $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

1.5 (or $\frac{3}{2}$) **[A1]**

(b) Correct vertical and horizontal change seen, e.g. $\frac{-3}{6}$ **[M1]**

-0.5 (or $-\frac{1}{2}$) **[A1]**

(c) $\frac{-9-7}{5-(-3)}$ **[M1]**

-2 **[A1]**

[Back to Question] *[Total Marks: 6]*

7 (a) (i) $\frac{3k-1-k}{5-2}$ **[M1]**

$\frac{2k-1}{3}$ **[A1]**

(ii) $\frac{2k-1}{3} = 4$ or $2k-1=12$ **[M1]**

6.5 or $\frac{13}{2}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

8 (a) $\frac{k-5}{4-(-2)} = -\frac{1}{2}$ **[M1]**

$k-5=-3$ **[M1]**

2 **[A1]**

(b) Gradient of $BC = \frac{-1-2}{10-4} = -\frac{1}{2}$ **[M1]**

Gradient of AB = gradient of BC and point B is common to both segments **[A1]**

(c) $\frac{(3p-1)-5}{p-(-2)} = 2$ **[M1]**

$3p-6=2(p+2)$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 8]*

9 (a) $\frac{(a+4)-3a}{5a-2a}$ or equivalent **[M1]**

$\frac{4-2a}{3a}$ **[A1]**

(b) States or implies gradient of PQ is $-\frac{1}{3}$ **[B1]**

$\frac{4-2a}{3a} = -\frac{1}{3}$ or $3(4-2a) = -3a$ **[M1]**

4 **[A1]**

(c) Gradient $AB = \frac{8-2}{3-(-1)}$ or $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

$\frac{23-8}{c-3} = \frac{3}{2}$ or $\frac{23-2}{c-(-1)} = \frac{3}{2}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 8]*

10 (a) $\frac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\frac{6-2a}{6}$ **[M1]**

$\frac{3-a}{3}$ or $1-\frac{a}{3}$ **[A1]**

(b) $\frac{3-a}{3} = 3$ oe **[M1]**

$a = -6$ **[A1]**

(c) $P = (-2,-18)$ and $Q = (4,0)$ **[M1]**

$\left(\frac{-2+4}{2}, \frac{-18+0}{2}\right)$ **[M1]**

$(1,-9)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

11 (a) $\frac{k-2}{4}$ **[B1]**

(b) Finds an expression for the gradient of BC (e.g., $\frac{k}{k-5}$) and equates to the gradient of AB: $\frac{k-2}{4} = \frac{k}{k-5}$ **[M1]**

Expands brackets correctly: $(k-2)(k-5) = 4k$

$k^2 - 7k + 10 = 4k$ [M1]

Rearranges to the given equation $k^2 - 11k + 10 = 0$ with no mathematical errors seen **[A1]**

(c) Factorises $(k-1)(k-10) = 0$ or uses the quadratic formula **[M1]**

1 and 10 **[A1]**

(d) Deduces that $k = 10$ must be used, as $k = 1$ would mean points A and C are the same, contradicting that they are distinct **[B1]**

Calculates the gradient of OB using $(5,10)$: $\frac{10-0}{5-0} = 2$ **[B1]**

[Back to Question] *[Total Marks: 8]*

12 (a) Calculates the gradient of AB as $\frac{9-3}{4-1} = 2$ and equates to gradient of BC or AC, e.g., $\frac{y-9}{8-4} = 2$ **[M1]**

$y = 17$ **[A1]**

(b) Sets up the gradient equation: $\dfrac{(p^2-p-3)-3}{p-1}=2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2-p-6=2(p-1)$

$\mathrm{p}^2-3p-4=0$ [M1]

Factorises $(p-4)(p+1)=0$ or uses the quadratic formula correctly **[M1]**

$p=4$ and $p=-1$ **[A1]**

(c) Sets up the gradient equation: $\dfrac{12-m}{(m+1)-3}=m$ **[M1]**

Multiplies by the denominator: $12-m=m(m-2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2-m-12=0$ **[M1]**

$m=4$ and $m=-3$ **[A1]**

[Back to Question] *[Total Marks: 10]*

13 (a) $\dfrac{5-1}{4-(-2)}$ or equivalent **[M1]**

$\dfrac{2}{3}$ **[A1]**

(b) $\dfrac{-2-k}{8-3}=-1.4$ or equivalent **[M1]**

$-2-k=-7$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

14 (a) $\dfrac{-9-5}{4-(-3)}$ or $\dfrac{5-(-9)}{-3-4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1-5}{k-(-3)}=-2$ oe **[M1]**

$-4=-2(k+3)$ or $-2k-6=-4$ **[M1]**

$k=-1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

15 (a) Forms a correct equation for the gradient: $\dfrac{5-2k}{2-(k-3)}=-3$ OR $\dfrac{2k-5}{k-3-2}=-3$ **[M1]**

Correctly simplifies the denominator and equates: $5-2k=-3(5-k)$ **[M1]**

$k=4$ **[A1]**

(b) Evaluates coordinates of P as $(1,8)$ and substitutes into midpoint formula:
$\left(\dfrac{1+2}{2},\dfrac{8+5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

16 **(a)** Finds a correct algebraic expression for the gradient: $\frac{8-4}{k-2-3k}$ or $\frac{4}{-2k-2}$ **[M1]**

Equates to $-\frac{1}{2}$ and eliminates fractions: $8 = 2k + 2$ or $-8 = -2k - 2$ **[M1]**

3 **[A1]**

(b) $P(9, 4)$ and $Q(1, 8)$ **[B1]**

(c) Identifies the coordinates of R must be of the form $(0, y)$ **[B1]**

Sets up the gradient equation: $\frac{y-4}{0-9} = 1$ **[M1]**

$(0, -5)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Calculates gradient of AB: $\frac{8-(-4)}{3-(-1)} = \frac{12}{4} = 3$ **[B1]**

Sets up equation for gradient involving C, e.g., $\frac{23-8}{p-3} = 3$ or $\frac{23-(-4)}{p-(-1)} = 3$ **[M1]**

8 **[A1]**

(b) Forms expression for gradient AD: $\frac{2q-1-(-4)}{q-(-1)}$ **[M1]**

Equates to $-\frac{1}{3}$ and simplifies: $3(2q + 3) = -1(q + 1)$ **[M1]**

$-\frac{10}{7}$ or -1.43 (accept $-1.428...$) **[A1]**

[Back to Question] *[Total Marks: 6]*

18 **(a)** Gradient $= \frac{(t^2 - 3t + 2) - 0}{t - 1}$ **[M1]**

Factorising numerator: $\frac{(t-1)(t-2)}{t-1}$ **[M1]**

Cancelling $(t - 1)$ correctly to obtain $t - 2$ **[A1]**

(b) $t - 2 = 2.8$ giving $t = 4.8$ **[M1]**

$y = 4.8^2 - 3(4.8) + 2$ **[M1]**

$(4.8, 10.64)$ **[A1]**

(c) Gradient of $QR = \frac{(t^2 - 3t + 2) - (-1.5)}{t - 3.2}$ **[M1]**

$\frac{t^2 - 3t + 3.5}{t - 3.2} = t - 2$ **[M1]**

$t^2 - 3t + 3.5 = t^2 - 5.2t + 6.4$ leading to $2.2t = 2.9$ **[M1]**

1.32 **[A1]**

[Back to Question] *[Total Marks: 10]*